

CC-112 Programming Fundamentals

Lab 05

1. Palindrome Number Check Using Recursion

Task: Write a recursive function isPalindrome(int n, int temp) that checks whether a given number is a palindrome.

Sample Input:

Enter a number: 121

Sample Output:

121 is a palindrome.

Sample Input:

Enter a number: 123

Sample Output:

123 is not a palindrome.

2. Prime Number Check Using a Function

Task: Write a function isPrime(int n) that returns 1 if n is prime and 0 otherwise.

Sample Input:

Enter a number: 29

Sample Output:

29 is a prime number.

Sample Input:

Enter a number: 15

Sample Output:

15 is not a prime number.

3. Reverse a Number Using Recursion

Task: Write a recursive function reverseNumber(int n, int rev) that returns the reverse of a given integer.

Sample Input:

Enter a number: 1234

Sample Output:

Reversed number: 4321

4. Sum of Digits Using Recursion

Task: Write a recursive function `sumOfDigits(int n)` that calculates and returns the sum of the digits of a given number.

Sample Input:

Enter a number: 567

Sample Output:

Sum of digits: 18

5. Exponentiation Using Recursion

Task: Write a recursive function `power(int base, int exp)` that calculates `base^exp`.

Sample Input:

Enter base: 2

Enter exponent: 5

Sample Output:

Result: 32

6. Predict the Output

```
#include <stdio.h>
```

```
void foo(int n) {  
    if (n <= 0)  
        return;  
    printf("%d ", n);  
    foo(n - 2);  
    printf("%d ", n);  
}
```

```
int main() {  
    foo(5);  
    return 0;  
}
```

Question: What will be the output of the above code?

7. Identify and Fix the Error

```
#include <stdio.h>
```

```
int findSquare(int num) {  
    int square = num * num;  
}
```

```
int main() {  
    int result = findSquare(5);  
    printf("Square: %d", result);  
    return 0;  
}
```

Question: Identify the error in the code and fix it.

8. Identify and Fix the Error

```
#include <stdio.h>
```

```
void add(int a, int b);
```

```
int main() {  
    float result = add(4, 5);  
    printf("Sum: %f", result);  
    return 0;  
}
```

```
void add(int a, int b) {  
    return a + b;  
}
```

Question: Identify the errors in the above code and rewrite the correct version.

9. Predict the Output

```
#include <stdio.h>
```

```
void changeValue(int x) {  
    x = 20;  
    printf("x : %d\n", x);  
}
```

```
int main() {  
    int x = 10;  
    printf("Before function call: %d\n", x);  
    changeValue(x);  
    printf("After function call: %d\n", x);  
    return 0;  
}
```

Question: What will be the output of the above code?